Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Platforms · Agentic Systems · RAG · Evaluation Pipelines · Python · TypeScript · LangGraph

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Apple Algorithms Programs — Sensing Technologies
Apple Inc. — Cupertino, California

Software Engineer — AI Platform (Algorithms)

Your job description names three things you are looking for: someone who built a chatbot that reasoned over internal docs, an agent that orchestrated multi-step tool calls, or a pipeline that turned unstructured data into searchable knowledge. I have built all three — not as tutorials, and not as prototypes. They are shipped, they are public, and engineers use them.

AgentProbe is the agent that orchestrated multi-step tool calls. I designed and shipped it end-to-end: a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with inter-agent state handoffs, tool calling, and human-in-the-loop validation via a React TypeScript observability dashboard. A **5-metric LLM-as-judge evaluation engine** runs on every commit — CI/CD regression gates block degraded runs before they reach production. Engineers get trace-level observability, per-run regression drift, and multi-provider LLM routing across OpenAI, Anthropic, Groq, and Ollama. It is live, open source, and running 100% locally at zero cost. github.com/venkatalakshmiparimala/AgentProbe

LeaseI is the pipeline that turned unstructured data into searchable, trusted knowledge. I built an agentic RAG pipeline over 1,240 residential lease PDFs spanning 5 US states and federal standards — extraction, chunking, embedding, semantic search via ChromaDB, and a **deterministic rules engine** that makes every compliance determination fully reproducible with **provenance at every step**. A pass@k evaluation harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring closes the feedback loop automatically. Result: **93.6% precision, 87.9% F1** on a corpus where ambiguity is the default and wrong answers have real consequences.

Before the AI projects, I spent 8 years at Entrata building production infrastructure other teams depended on daily — **20+ full-stack modules, 1M+ records, 100K+ monthly active users, 99.5% uptime**. I led distributed-systems redesigns, built access-controlled data pipelines with full PHI compliance (zero incidents), and shipped the company's first production AI pipeline on 1.6M real financial records with quality gates, output validation, and bias-aware compliance checks. Working cross-functionally across compliance, operations, finance, and product teams for 120+ Agile sprints taught me the thing that matters most in internal platforms: the tool only compounds in value if the teams around you trust it completely.

The work you described — AI platforms connecting engineers to institutional knowledge, agentic pipelines with human-in-the-loop validation, CI-integrated quality monitoring, provenance-backed reproducible analysis — is work I have already done, at production quality, in public. I would bring that infrastructure and that standard to Apple's algorithm teams.

Thank you for your time. I would welcome the chance to show you what I have built.

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